NW London Outpatient Pathways

Thyroid dysfunction in pregnancy

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Monitoring:

Ideally measure TFTs:
At pregnancy confirmation
At antenatal booking
At least once in 2nd and 3rd trimester
2-4 weeks post-partum

New diagnosis: Hypo / Hyperthyroidism

Contact endocrinology for advice & guidance

2WW Referral Criteria

2WW referral if patients presenting with thyroid nodule or goitre and suspected malignancy

Overt or subclinical Hypothyroidism

Urgent referral to antenatal and endocrinology

Check TSH and FT4 immediately and discuss with specialist whilst patient is awaiting assessment

On specialist advice:
Whilst awaiting review, discuss initiation of levothyroxine or if already taking discuss changes to current dosage and monitoring requirements.
NB> an increase in the levothroyxine dose may be required (by 30-50%) from as early as 4-6 weeks gestation to maintain normal serum TSH levels

Postpartum
If a patient was taking levothyroxine during pregnancy **discuss with specialist** if any changes to the dose are required and the frequency of TFT monitoring
If patient was taking levothyroxine prior to pregnancy the dose will need to be decreased back to pre-pregnancy dose **as advised by a specialist**. Check TFTs at 6 weeks postpartum

Overt or subclinical Hyperthyroidism

Emergency Admission Criteria

Arrange emergency admission if the woman has severe signs and symptoms of hyperthyroidism (e.g. thyroid storm) or intractable vomiting suggesting hypermesis gravidarum.

Urgent referral all patients to antenatal and endocrinology including women who are euthyroid

Seek advice from an endocrinologist:
Patient has adrenergic symptoms (e.g. tremor or tachycardia) which requires symptomatic treatment whilst awaiting specialist assessment

Check TSH, FT3 and FT4 immediately and send results to specialist

Specialist Treatment
If the patient is already taking anti-thyroid treatment, ongoing blood monitoring should be arranged and managed in secondary care.
Propylthiouracil is usually used in pregnant women due to the increase possible risk of congenital abnormalities with carbimazole therapy (see MHRA alert) The lowest possible dose should be used to maintain euthyroidism.
NB: The potential risks vs benefits of individual anti-thyroid treatments should be considered before prescribing in pregnancy.

Postpartum
There is a significant risk of relapse Check TFTs after delivery. Check TSH and FT4 6-8 weeks postpartum, particularly in patients with the following: Goitre, non-specific symptoms suggestive of thyroiditis, history of PPT or autoimmune thyroid disease or TPOAb positive

Refer to endocrinology if TFT results are abnormal (urgency according to clinical judgement)

Postpartum thyroiditis (PPT) present

PPT usually resolves spontaneously within 1 year postpartum.

Thyrotoxic pattern
If initial TFTs show thyrotoxic pattern – **Refer to a specialist** for further tests to differentiate from Grave's disease. After resolution of thyrotoxic phase, monitor TFTs after 4-8 weeks (or if new symptoms develop) to screen for hypothyroid phase.

Hypothyroid pattern
If TFTs indicate a hypothyroid pattern, **seek advice from specialist** on whether to start levothroxine treatment.

History of PPT (which has resolved)
- Offer annual TFT monitoring
- Screen women prior to future pregnancy
- 6-8 weeks after pregnancy

(Due to high risk of permanent hypothyroidism and recurrence in subsequent pregnancies).

Women with T1 Diabetes
3 x more at risk of PPT. Therefore, TSH, FT4 and TPOAb status should be checked:
- Prior to pregnancy
- At diagnosis of pregnancy
- 3 and 6 months postpartum

Undetected subclinical hypothyroidism:
During pregnancy, this may adversely effect neuropsychological development and survival of the foetus. Associate ovulatory dysfunction and infertility.

If maternal hyperthroidism is inadequately controlled: This may increase risk of miscarriage, pre-eclampsia, intrauterine growth restriction, pre-term delivery, low birth weight, fetal death

Thyroid dysfunction in pregnancy
- ↑ oestrogen production and TBG concentrations lead to ↑ FT4 and FT3.
- Fall in TSH is normal in the 1st trimester.

Trimester-related reference ranges should be applied for TSH and for total and free thyroid hormones (FT4 and FT3).
Alterations in immune function occur, which can influence course of existing autoimmune thyroid disease.
After delivery, levels of thyroid hormones and TSH normally return to the pre-pregnant state.